



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

Y. Sravan Kumar (192472289)

Problem Solving Using Numpy, Pandas, Scipy, Scikit Learn Libraries in Python for Data Scientist

1. GENERATE RAW MOCK DATA (Simulating an imperfect CSV file)

Code:

```
employee_data.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/employee_data.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# 1. GENERATE RAW MOCK DATA (Simulating an imperfect CSV file)
raw_data = {
    "Employee_Name": [" Alice ", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, np.nan, 92000, 60000], # Missing value
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28",
    ],
}

# Create DataFrame
df = pd.DataFrame(raw_data)

# Display Raw Data
print("--- Raw Data ---")
print(df)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/employee_data.py
--- Raw Data ---
Employee_Name  Department  Salary  Join_Date
0      Alice           HR    50000.0  2022-01-15
1         Bob  Engineering    85000.0  2021-06-20
2      Charlie           HR         NaN  2023-03-11
3      Diana  Engineering    92000.0  2020-11-01
4         Evan  Marketing    60000.0  2024-02-28
>>>
```

2. DATA PREPROCESSING

Code:

```
data_preprocessing.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/data_preprocessing.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

# Employee Data
raw_data = {
    "Employee_Name": [" Alice ", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, np.nan, 92000, 60000],
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28"
    ]
}

df = pd.DataFrame(raw_data)

# 2. DATA PREPROCESSING

# Remove extra spaces from Employee Name
df["Employee_Name"] = df["Employee_Name"].str.strip()

# Replace missing salary with the median salary
median_salary = df["Salary"].median()
df["Salary"] = df["Salary"].fillna(median_salary)

# Convert Join_Date to datetime format
df["Join_Date"] = pd.to_datetime(df["Join_Date"])

# Create Years_of_Service column
df["Years_of_Service"] = 2026 - df["Join_Date"].dt.year

print("---- Preprocessed Data ----")
print(df)
|
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/data_preprocessing.py
--- Preprocessed Data ---
Employee_Name Department Salary Join_Date Years_of_Service
0 Alice HR 50000.0 2022-01-15 4
1 Bob Engineering 85000.0 2021-06-20 5
2 Charlie HR 72500.0 2023-03-11 3
3 Diana Engineering 92000.0 2020-11-01 6
4 Evan Marketing 60000.0 2024-02-28 2
>>>
```

3. DATA AGGREGATION

Code:

```
DATA AGGREGATION.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/DATA AGGREGATION.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

# Employee Data
raw_data = {
    "Employee_Name": ["Alice", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, 72500, 92000, 60000],
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28"
    ]
}

df = pd.DataFrame(raw_data)

# 3. DATA AGGREGATION
# Group by Department and calculate average salary and total employees
dept_summary = (
    df.groupby("Department")
    .agg(
        Avg_Salary=("Salary", "mean"),
        Total_Employees=("Employee_Name", "count")
    )
    .reset_index()
)

print("--- Aggregated Summary ---")
print(dept_summary)
```

Output:

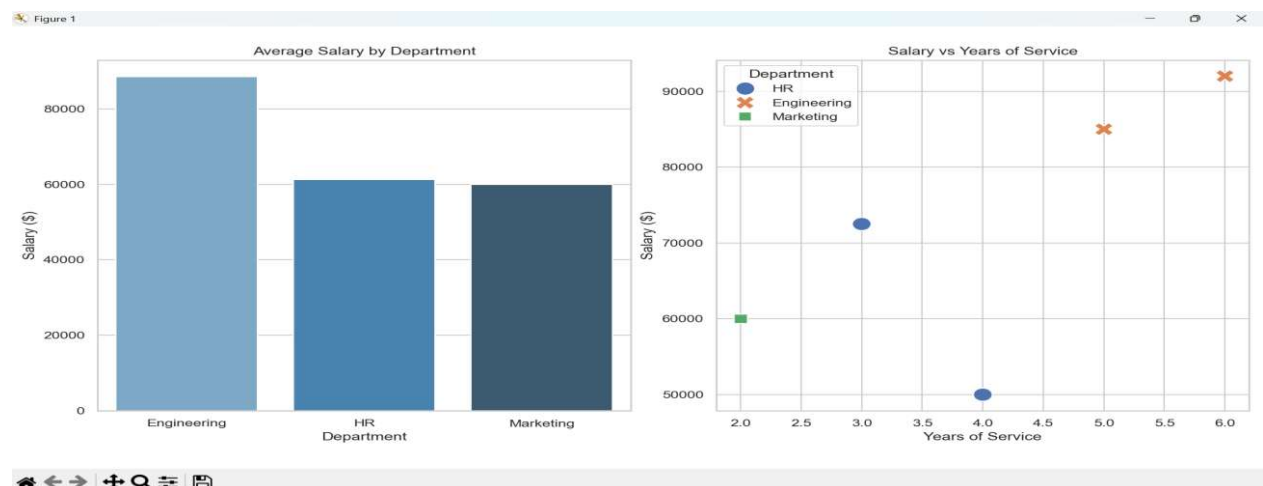
```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/data_preprocessing.py
--- Preprocessed Data ---
Employee_Name Department Salary Join_Date Years_of_Service
0 Alice HR 50000.0 2022-01-15 4
1 Bob Engineering 85000.0 2021-06-20 5
2 Charlie HR 72500.0 2023-03-11 3
3 Diana Engineering 92000.0 2020-11-01 6
4 Evan Marketing 60000.0 2024-02-28 2
>>>
```

4. DATA VISUALIZATION

Code:

```
DATA VISUALIZATION.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/DATA VISUALIZATION.py (3.12.3)
File Edit Format Run Options Window Help
)
    .reset_index()
)
# 4. DATA VISUALIZATION
# Set theme
sns.set_theme(style="whitegrid")
# Create two plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
# Plot 1: Bar Chart
sns.barplot(
    data=dept_summary,
    x="Department",
    y="Avg_Salary",
    hue="Department",
    palette="Blues_d",
    legend=False,
    ax=axes[0]
)
axes[0].set_title("Average Salary by Department")
axes[0].set_xlabel("Department")
axes[0].set_ylabel("Salary ($)")
# Plot 2: Scatter Plot
sns.scatterplot(
    data=df,
    x="Years_of_Service",
    y="Salary",
    hue="Department",
    style="Department",
    s=200,
    ax=axes[1]
)
axes[1].set_title("Salary vs Years of Service")
axes[1].set_xlabel("Years of Service")
axes[1].set_ylabel("Salary ($)")
plt.tight_layout()
plt.show()
```

Output:



Creating Sample Data Frame

Code:

```
Dataframe.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Dataframe.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd

# Create an empty DataFrame
df = pd.DataFrame()

# Add columns
df['Name'] = ['John', 'Emma', 'Liam', 'Olivia']
df['Age'] = [20, 19, 21, 18]
df['Student'] = [True, True, False, True]

# Display the DataFrame
print(df)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Dataframe.py
      Name  Age  Student
0   John   20     True
1  Emma   19     True
2   Liam   21     False
3 Olivia   18     True
>>>
```

Adding Row to Data Frame

Code:

```
adding row to dataframe.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/adding row to dataframe.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd

# Create the DataFrame
df = pd.DataFrame()

df['Name'] = ['John', 'Emma', 'Liam', 'Olivia']
df['Age'] = [20, 19, 21, 18]
df['Student'] = [True, True, False, True]

# Create a new row
new_row = pd.DataFrame(
    [['Sophia', 22, False]],
    columns=['Name', 'Age', 'Student']
)

# Add the new row to the DataFrame
df = pd.concat([df, new_row], ignore_index=True)

# Display the updated DataFrame
print(df)
```

Output:


```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/adding row to dataframe.py
   Name Age Student
0  John  20     True
1  Emma  19     True
2  Liam  21    False
3 Olivia  18     True
4 Sophia 22    False
>>>
```

Data Analysis with SciPy

1.Importing Required Libraries

Code:

```
addinglibraries.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\addinglibraries.py (3.12.3)
File Edit Format Run Options Window Help
import numpy as np
import pandas as pd
from scipy import stats

print("Libraries imported successfully.")
import numpy as np
from scipy import stats

data = [10, 20, 30, 40, 50]

print("Data:", data)
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data, keepdims=True))
```

2.Measures of Central Tendency Using SciPy

Code:

```
addinglibraries.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\addinglibraries.py (3.12.3)
File Edit Format Run Options Window Help
import numpy as np
import pandas as pd
from scipy import stats

print("Libraries imported successfully.")
import numpy as np
from scipy import stats

data = [10, 20, 30, 40, 50]

print("Data:", data)
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data, keepdims=True))
```

3.Probability Distribution Analysis Using SciPy

Code:

```
pda using scipy.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/pda using scipy.py (3.12.3)
File Edit Format Run Options Window Help
from scipy.stats import norm

probability = norm.cdf(85, loc=70, scale=10)

print("Probability:", round(probability, 4))
```

4.Hypothesis Testing

Code:

```
hypothesistesting.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/hypothesistesting.py (3.12.3)
File Edit Format Run Options Window Help
from scipy import stats

data = [22, 25, 19, 24, 28, 30]

t_stat, p_value = stats.ttest_1samp(data, 25)

print("T-Statistic:", round(t_stat, 3))
print("P-Value:", round(p_value, 3))
```

5.Correlation Analysis

Code:

```
correlational analysis.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/correlational analysis.py (3.12.3)
File Edit Format Run Options Window Help
from scipy.stats import pearsonr

x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

corr, p_value = pearsonr(x, y)

print("Correlation:", corr)
print("P-Value:", p_value)
```

6. Linear Algebra Operations

Code:

```
Linear operation.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Linear operation.py (3.12.3)
File Edit Format Run Options Window Help
from scipy import linalg

A = [[3, 2],
     [1, 2]]

B = [5, 5]

solution = linalg.solve(A, B)

print("Solution:", solution)
```

7.Optimization Using SciPy

Code:

```
optimization using scipy.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/optimization using scipy.py (3.12.3)
File Edit Format Run Options Window Help
from scipy.optimize import minimize

def objective(x):
    return x**2 + 4

result = minimize(objective, x0=5)

print("Optimal Value of x:", result.x)
print("Minimum Function Value:", result.fun)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/addinglibraries.py
Libraries imported successfully.
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/addinglibraries.py
Libraries imported successfully.
Data: [10, 20, 30, 40, 50]
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=array([10]), count=array([1]))
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/pda using scipy.py
Probability: 0.9332
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/hypothesistesting.py
T-Statistic: -0.205
P-Value: 0.846
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/correlational analysis.py
Correlation: 1.0
P-Value: 0.0
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Linear operation.py
Solution: [0. 2.5]
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/optimization using scipy.py
Optimal Value of x: [-2.62955131e-08]
Minimum Function Value: 4.000000000000001
>>>
```

Scikit Library in Python

1. Logistic Regression (Supervised Learning)

Code:

LR.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/LR.py (3.12.3)

File Edit Format Run Options Window Help

```
import numpy as np
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Load Iris dataset
iris = datasets.load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Standardize the data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Train Logistic Regression model
log_reg = LogisticRegression()

log_reg.fit(X_train, y_train)

# Predict test data
y_pred = log_reg.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

Output:

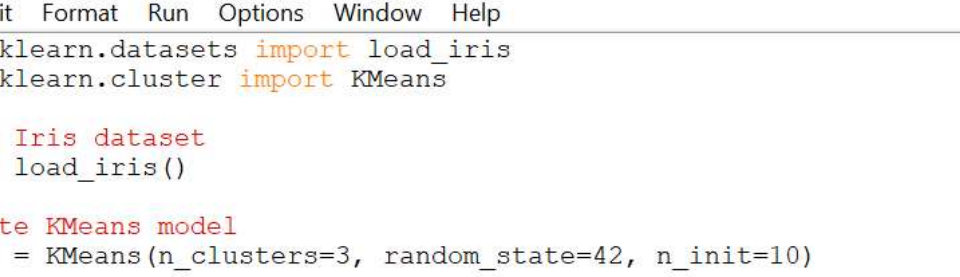
IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/LR.py
Accuracy: 1.0
>>>
```

2. K Means Clustering (Unsupervised Learning)

Code:



The screenshot shows a text editor window with a menu bar (File, Edit, Format, Run, Options, Window, Help) and a code editor. The code is a Python script for KMeans clustering using sklearn. It imports load_iris and KMeans, loads the Iris dataset, creates a KMeans model with 3 clusters, trains it, and prints the cluster labels.

```
cluistering.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/cluistering.py (3.12.3)
File Edit Format Run Options Window Help
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans

# Load Iris dataset
iris = load_iris()

# Create KMeans model
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

# Train the model
kmeans.fit(iris.data)

# Get cluster labels
cluster_labels = kmeans.labels_

print("Cluster Labels:")
print(cluster_labels)
```

Output:

[illegible]

